

Kemal Baran Dursun

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EDUCATION

McGill University

Montreal, QC

Bachelor of Arts - Double Major in Computer Science & Economics | GPA: 3.44/4.0

Expected December 2026

- Relevant Coursework: Algorithms & Data Structures, Software Systems, Computer Systems, Programming Languages & Paradigms, Theory of Computation, Programming Challenges, Probability, Econometrics

TECHNICAL SKILLS

Languages: TypeScript, Python, SQL, Bash, Java

Frameworks & APIs: React, React Native, Node.js/Express, FastAPI, Hono, REST APIs

Data & DevOps: PostgreSQL, Prisma, Redis, Supabase, Docker, Railway, Git, GitHub Actions, Vitest

ML & Analytics: XGBoost, statsmodels, scikit-learn

PROJECTS

FragMance | React Native, TypeScript, Node.js/Express, PostgreSQL, Redis | [App Store](#)

- Built and launched FragMance on the Apple App Store as sole engineer, delivering a 21-screen React Native app, a 58-endpoint Node.js/Express REST API, and an automated deployment pipeline.
- Engineered typo-tolerant search across 63,000 fragrances using PostgreSQL trigram indexes and confidence ranking; developed content-based recommendations using IDF-weighted fragrance-note similarity.
- Established automated quality checks with 500+ tests and GitHub Actions CI; identified and fixed a production PII-exposure vulnerability by correcting database access controls and adding regression tests.

Recap Engine | TypeScript, Hono, React 19, PostgreSQL, Docker | [Live Demo](#)

- Deployed an episode-aware TV recap app with a Hono API, React frontend, and PostgreSQL database; containerized the application with Docker for Railway hosting.
- Designed a shared recap cache keyed by show and episode boundary; cache hits incur \$0 additional model cost and consume no quota, with automated tests verifying that repeat requests trigger no new generation.
- Secured user access with Google OAuth, hashed server sessions, CSRF protection, and encrypted API-key storage; tested cross-user isolation.
- Integrated a Manifest V3 Chrome extension to detect episodes on Netflix, Prime Video, and Disney+; maintained an automated Vitest test suite.

EconSight | Python, FastAPI, PostgreSQL, XGBoost, React | [Live Demo](#)

- Built a Canadian macroeconomic forecasting platform that ingests 9 Statistics Canada and Bank of Canada indicators into a PostgreSQL database through a raw → cleaned → model-ready pipeline.
- Served time-series and XGBoost forecasts through a FastAPI backend, with uncertainty ranges and feature-importance explanations displayed in a React dashboard.
- Added a Q&A feature that retrieves relevant data and passes it to an LLM (Llama 3.3) to answer questions about the indicators, with downloadable PDF reports.

PROFESSIONAL EXPERIENCE

Advanced Information Modeling Inc

St. Petersburg, FL (Remote)

Software & IT Systems Intern

June 2026 - August 2026

- Built a Python script to parse application log files and group recurring errors by type, making it faster for the team to spot repeat crashes.
- Queried and summarized system data using SQL for weekly reports and team reviews.
- Developed automated Python tests to verify data-import outputs and invalid-input handling before each production release.

Duronm Textile Company

Izmir, Turkey

Data & Business Systems Intern

June 2025 - August 2025

- Extracted textile order and billing data using SQL for the finance team's weekly reports.
- Created Excel reports with pivot tables and lookups to track invoices and payments, replacing manual copy-paste tracking.
- Automated order-data preparation with Python scripts to clean and combine exported datasets before analysis.

Languages: English (Fluent), Turkish (Native), French (Intermediate)